

The following report evaluates World Vision Australia's Bounce Back campaign, a fundraising program that engages existing child sponsors through milestone appeals such as Birthday, Christmas, and Education. While the campaign has maintained high response rates, overall donation values have steadily declined from \$21M in 2018 to \$1M in 2025. The campaign plays a vital role in strengthening donor relationships and generating additional support, analysis shows that its current design is misaligned with donor behaviour. Heavy reliance on email, uniform ask strategies, and timing driven by internal production schedules have led to inefficiencies, campaign fatigue, and a decline in overall donor value.

Descriptive and predictive analysis confirmed that tenure is the strongest driver of giving capacity, with older and long-tenure supporters consistently donating at higher levels. Donors aged over 65 contribute the highest average donation amounts, with an overall average of \$31.32 per gift. Similarly, long-tenured supporters (50+ years) represent the most generous segment, averaging \$55.72 per donation. However, these high-value donors are often channelled through email-first pathways that underperform compared to Direct Mail and Telephone. Timing analysis further revealed that communication peaks do not align with donor readiness, with donations clustering around mid-month but outbound messages concentrated earlier and later in the cycle. Finally, the reliance on uniform ask ladders suppresses upgrades, overstretches newer donors, and underutilises the generosity of loyal supporters, while over-reliance on Christmas and Education appeals limits the potential of Birthday campaigns.

To address these challenges, the report recommends:

1. **Rebalance channel allocation** – Shift 20% of outreach for older/long-tenure donors into Direct Mail and Telephone, while maintaining digital efficiency for younger groups.
2. **Optimise timing around donor readiness** – Concentrate key asks between Days 13-21, with lighter, emotionally framed messages after Day 21 to reduce fatigue.
3. **Personalise ask framing** – Anchor suggested donation amounts to donor capacity and tenure, using realistic increments and milestone framing to encourage upgrades.

Implementing these strategies will reduce wasted communications, protect supporter goodwill, and unlock higher-value gifts from loyal donors. By aligning channels, timing, and ask strategies with donor behaviour, WVA can strengthen revenue efficiency and ensure the long-term sustainability of the Bounce Back campaign.

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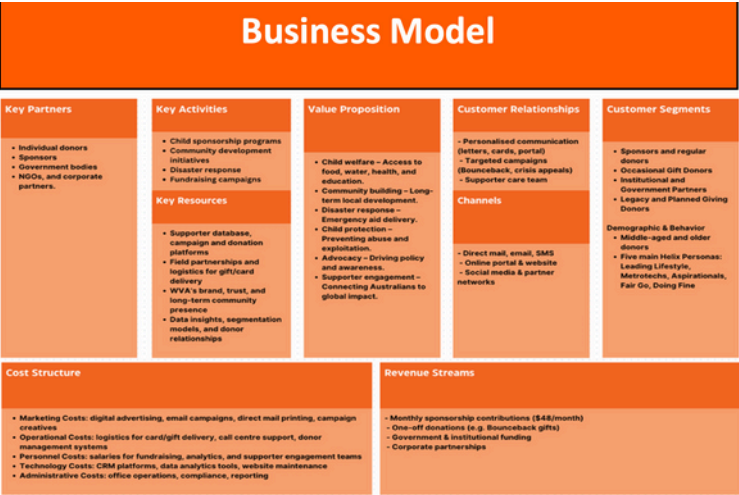
Introduction

Business Context

World Vision Australia is a not-for-profit organization that partners with communities, donors, and governments to improve child wellbeing through long-term development programs. It is part of the global World Vision Partnership operating in over 100 countries with 35,000 staff. WVA's mission is to empower Australians to create lasting change for vulnerable children and their communities.

KPI for FY23

Sponsored 175,697 children	364 Projects in 41 Countries	Raised \$442.7 million
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Bounce Back Campaign

In support of this mission, WVA launched the Bounce Back campaign in 2018 as a tactical fundraising initiative designed to generate additional one-off donations from existing child sponsors. These milestone-based communications such as birthday cards, Christmas greetings, and education-focused messages aim to strengthen emotional connection while enabling more flexible funding beyond regular monthly pledges.

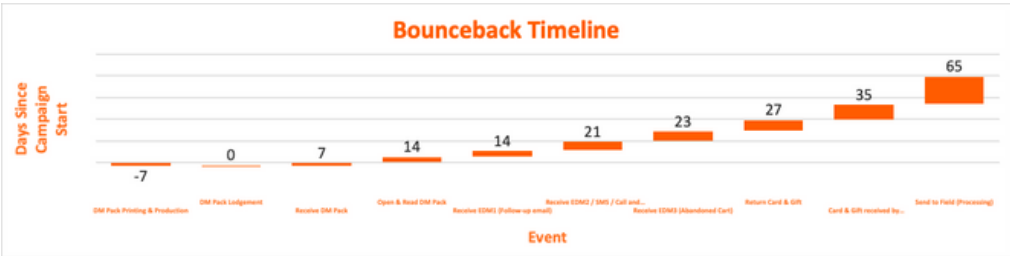


Figure 1: BounceBack Campaign Timeline by Touchpoints

Problem Statement

Supporters across different demographic groups are receiving communications at suboptimal times and through mismatched channels, leading to lower engagement and inefficient donation outcomes. To improve ROI and maximise donation value, we need to understand how communication sequence, timing, and channel impact supporter response throughout the BounceBack campaign.

Approach

To address these challenges, the analysis proceeds in three stages:

- Descriptive analysis to identify key trends in donations, channel performance, and timing patterns across campaigns
- Diagnostic analysis to examine underlying causes of declining effectiveness, including channel misalignment, over-communication, and timing inefficiencies
- Predictive modelling to test the influence of campaign factors such as communication sequence, channel allocation, and ask strategy, identifying the strongest drivers of supporter response.

Together, these steps provide the foundation for targeted recommendations to rebalance channels, align timing with donor readiness, and optimise ask amounts to improve campaign ROI and donor engagement

Assumption

- Analysis emphasized interaction data, donation, and supporter groupings data tables to derive insights supporting the problem statement.
- All negative figures (e.g., donation amounts, tenure anomalies) were considered outliers and excluded from the analysis to ensure data integrity.
- Predictive models used four donation categories: Non-donor = 0 or less, Low = < 10, Medium = ≤ 50, High = > 50.

Descriptive Analysis

Overall trend

Donation amounts have shown fluctuations, with early peaks followed by a consistent decline. The highest donation value was recorded in 2018 at \$21 million, which gradually decreased to \$13 million by 2024, before dropping sharply to just \$1 million in 2025. Throughout this period, the response rate remained stable at around 99%, with only a slight dip in 2023. Overall, while donors continue to respond, the value of donations has been shrinking steadily over time.

Donors Profile

Analysis of donor demographics highlights clear patterns in giving behavior. Older donors (65+) contribute the highest average donation amounts, with an overall average of \$31.32 per gift. Similarly, long-tenured supporters (50+ years) represent the most generous segment, averaging \$55.72 per donation. Communication channel analysis also reveals a strong contrast between physical and digital methods. Letters (\$90) and Telephone (\$74) generate significantly higher donation values compared to digital channels, such as Web (\$31.22). This demonstrates the higher value potential of older, long-term supporters and traditional communication channels..

Communication Chanel and Sequence

The distribution of communication channels shows clear imbalances. Email dominates, accounting for 72% of all communications, while Direct Mail contributes 18% and SMS 10%. Despite this heavy reliance on email, the most effective donation outcomes are observed from early EDMs and Direct Mail (over \$6 million combined). By contrast, late EDMs and SMS produce the weakest results (around \$0.8 million). This suggests that while email volume is high, it is associated with diminishing returns, whereas Direct Mail, although underused, demonstrates stronger value potential.

Timing

Donation behavior follows a cyclical pattern aligned with communication pushes. Donations typically rise and fall in waves, with the largest peak occurring around Day 15, when donations reached \$2 million. After this high point, donations declined mid-cycle, followed by smaller surges before tapering off at the end of the month. This indicates that donations closely mirror communication intensity, with Day 15 emerging as the strongest giving point.

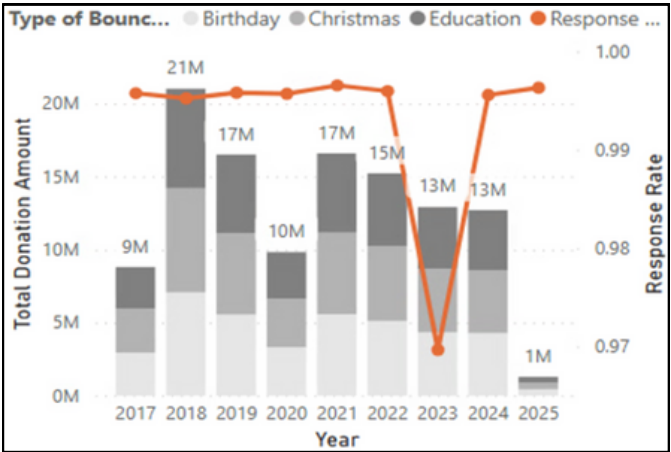


Figure 2. Total Donation Amount and Response Rate by Year and Bounceback Type

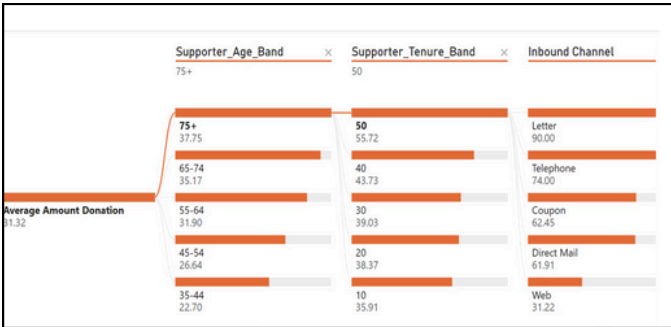


Figure 3: Tree chart of Age Band Supporter Tenure with Preferred Inbound Channel

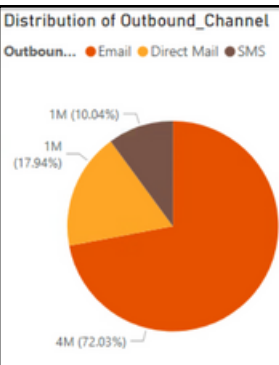


Figure 4. Distribution of Outbound Channel (Aggregate Contribution)

Bounceback channel and sequence	Total communication sent	Total Donat received
Abandoned	48788	1064838
CBB FY18 - Online	1854	16982
Chaser - Bethlehem		
DM	1214589	615456
eDM1	2000769	3443602
eDM2	1573071	3456986
eDM3	274941	749069
Resend	488887	1355798
SMS1	575141	1287468
SMS2	16679	53617
Total	6194719	38808174

Figure 5. Sequence Performance (Total Communications vs Total Donations)

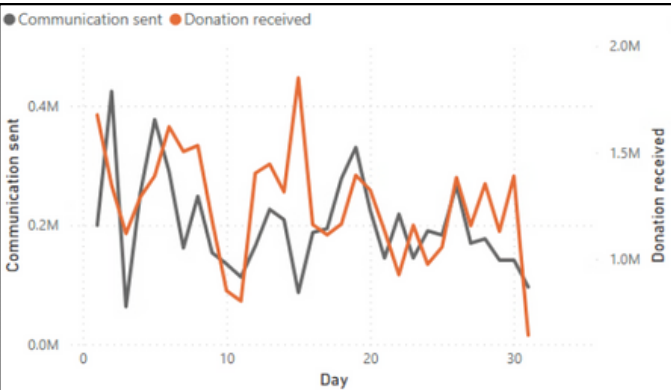


Figure 6. Communications Sent and Donations Received by Day

Diagnostic Analysis

Q1. Are we using the right channels for the right supporters?

There is a structural misalignment between channel allocation and donor preference. Outbound is dominated by Email (72%) while Direct Mail and SMS are underused (Figure 5). Yet high-value inbound channels such as Letter (\$90) and Telephone (\$74) (Figure 2) consistently outperform digital paths. Besides, Figure 6 also shows older/long-tenure supporters (55+) give \$45-60 via DM/Face-to-Face, compared to only \$25 through Web or Email. Even younger donors (<45) lean higher on Telephone (\$35) and DM (\$29) than Email/Web (\$17-25) (Figure 7). Despite this evidence, campaigns still default to uniform Email→Web journeys, suppressing ROI, leaving higher-value contributions untapped, and eroding supporter goodwill due to mis-matched channel experiences.

Bounceback_channel_and_sequence	Sum of TotalPaid	Response_InboundChannelName	Average of TotalPaid
Abandoned	1309984	Telephone	35.36
CBB FY18 - Online Chaser - Bethlehem	52664	Direct Mail	29.32
DM	4950354	Coupon	24.72
eDM1	5167718	Unknown	21.42
eDM2	5147691	Email	17.65
eDM3	2648517	Web	16.86
Resend	3837004	Face to Face	15.00
SMS1	4779131	Live Chat	15.00
SMS2	789978	Letter	11.43
		SMS	10.00
Total	5456079	Total	21.48

Figure 7. Sequence Performance (Over 55 Age Band Supporters)

Bounceback_channel_and_sequence	Sum of TotalPaid	Response_InboundChannelName	Average of TotalPaid
Abandoned	5769161	Face to Face	61.56
CBB FY18 - Online Chaser - Bethlehem	295921	Direct Mail	45.50
DM	29959578	SMS	42.00
eDM1	26424893	Coupon	36.52
eDM2	26135849	Letter	34.23
eDM3	9687716	Telephone	33.14
Resend	16757004	Live Chat	30.69
SMS1	22342093	Unknown	27.68
SMS2	3429503	Web	25.29
		Email	25.20
Total	31328115	Total	33.52

Figure 8. Sequence Performance (Under 45 Age Band Supporters)

Q2. Are we communicating when donors are actually ready to give?

We identified a misalignment between supporter readiness and campaign cadence. Donations peak around Day 15 (Figure 3), yet outbound activity is concentrated too early in Week 1 and again late in the cycle (Days 18-20), leaving the mid-cycle window under-served. This pattern is reinforced by structural drivers: Direct Mail lodgment on the 28th anchors EDM/SMS follow-ups into late bursts, while batch-sending habits and print lead-times prevent alignment with donor intent. Donation latency confirms that conversions cluster 10-14 days post-DM, but current scheduling delivers before intent has matured or after momentum has decayed. Without rebalancing effort toward Days 13-15 and setting guardrails to limit post-Day 20 volume, WVA risks missing peak responsiveness and compounding fatigue through poorly timed contact.

Q3. Are we over-communicating and reducing efficiency?

Analysis shows that the majority of value is realized in the first three touches, with DM, EDM1 & EDM2 delivering the strongest returns. Beyond this point, returns collapse: EDM3 and SMS2 contribute negligible incremental revenue despite large send volumes (Figure 5). This pattern reflects a structural inefficiency in late-stage outreach, where effort continues but yield falls away. By contrast, intent-led steps such as Resend and Abandoned Cart are significantly more efficient, demonstrating that aligning contact to supporter behavior drives higher returns. Persisting with blanket late-cycle pushes risks wasted spend and increased opt-outs without delivering meaningful lift.

Q4. Are we asking for the right amount?

The decline in donor engagement is driven by campaigns that are not personalised to donor history or supporter profiles, with crowded sequencing and repetitive messaging leading to disengagement. Over-reliance on Christmas and Education campaigns has created diminishing returns, while Birthday remains under-leveraged. The current ask strategy relies on fixed multipliers of past giving (1.25x, 1.5x, 2x) for responders and a flat \$20–45 ladder for non-responders and new supporters. This ignore natural differences in donor capacity: younger donors (<45) typically give \$20–30, older donors (55+) exceed \$60 through DM and phone completions (Figure 7,8), and tenure further distinguishes potential, with <40 years donating up to \$200 and >40 years often reaching \$500 (Appendix 3). This lack of differentiation has overstretched newer donors, under-asked loyal supporters, and ultimately constrained upgrades and weakened overall conversion.

Predictive Analysis

We applied **Random Forest models** to each BounceBack campaigns (Birthday, Christmas, Education) to test prediction-focused hypotheses about donation outcomes, with performance evaluated at the ensemble level. For interpretability, we extracted representative decision trees (Figure 9) from the forest to surface practical rules around tenure thresholds, channel effects, and ask ladder sensitivity. Outcomes were categorized into three value bands: **Low (\$1-\$10)**, **Medium (\$11-\$50)**, and **High (\$51+)**.

Decision trees were chosen because their if-then rules can be translated directly into operational journey changes, such as adjusting channels, sequencing, or ask ladders. Models were validated using cross-validation and standard classification metrics.

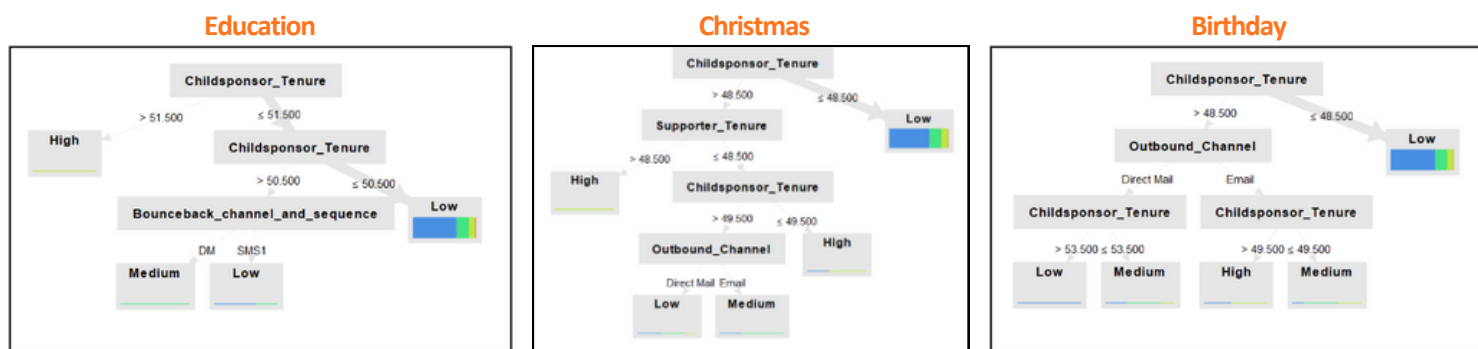


Figure 9. Decision Tree based on Campaigns

Hypothesis and Findings

Hypothesis 1: Longer donor tenure significantly increases the likelihood of Medium and High donations.

Descriptive analysis showed that longer sponsorship tenure is linked to higher giving levels, indicating that tenure influences donation outcomes. If this relationship holds consistently, supporter journeys can be segmented by tenure band to maximize ROI.

Findings:

- Tenure was the root split in all three campaigns. Thresholds varied slightly (Birthday 48.5 yrs, Christmas 48.5 yrs, Education 51.5 yrs), but the pattern was consistent: longer-tenure supporters were more likely to donate at higher levels.
- A plateau effect was observed in the Birthday campaign: uplift occurred around 49-53 years, but beyond 53 years the predicted outcome dropped back to Low, suggesting diminishing returns for extremely long-tenure supporters.

Conclusion: Tenure is a reliable predictor of donor value, but segmentation should avoid over-indexing on very long-tenure supporters, where incremental gains flatten.

Hypothesis 2: Offline channels are more effective than online.

Descriptive analysis suggested that Direct Mail and Telephone consistently deliver higher average gifts than Email or SMS, particularly among older and long-tenure supporters. However, predictive results show that channel performance is not absolute - it depends on campaign context.

Findings:

- **Birthday:** Direct Mail lifted some long-tenure donors into Medium outcomes, though overall impact plateaued quickly.
- **Christmas:** High gifts were tenure-driven, but Email outperformed Direct Mail for mid-tenure supporters, reflecting seasonal responsiveness to faster digital channels.
- **Education:** Donations were tenure-driven; Direct Mail added limited uplift into Medium gifts, while SMS remained stuck at Low.

Conclusion: Offline channels remain the strongest drivers of value, but their advantage is campaign-sensitive. A one-size-fits-all channel strategy risks missed opportunities; tailoring channel mix to both supporter profile and campaign type will maximize ROI.

Hypothesis 3: Donor response to ask levels varies by tenure and giving history.

Descriptive analysis of gift distributions showed strong clustering around \$10, with secondary peaks at \$20, \$40, and \$50, and smaller groups at higher levels (\$90, \$100, \$200+). If donor sensitivity to ask amounts aligns with tenure and past giving, tailored ask ladders should improve conversion compared to flat, channel-driven asks.

Findings:

- Most Low donors default to \$10 regardless of channel or campaign, showing limited stretch potential.
- Mid-value donors clustered at \$20-\$50 responded better to incremental steps that match these natural peaks.
- Current uniform ladders (e.g., \$20/\$35/\$45) introduce “unnatural” amounts that are not supported by distribution data, creating risk of over-asking.

Conclusion: Ask sensitivity is not uniform - it depends heavily on tenure and prior gift history. Aligning ask handles with observed giving peaks (e.g., \$10 → \$15 → \$20 for Low donors, \$20 → \$40 → \$50 for Medium, \$75+ for High) is more effective than applying generic ladders across all segments.

Model Performance

The Random Forest models achieved reasonable accuracy in classifying Low donors, but struggled to reliably distinguish Medium and High outcomes. This is reflected in a **Kappa 0** (Appendix 2), indicating that while the model confirms Low-value donors, it offers limited incremental reliability beyond chance when predicting higher-value gifts. As such, the trees are best interpreted as decision-support tools for surfacing practical segmentation rules (e.g., tenure thresholds, channel splits) rather than as standalone prediction engines.

Interpretation

These insights show that WVA's revenue challenges stem less from lack of donor generosity and more from a misalignment of campaign design with behavioral evidence. The findings reinforce that tenure is the most powerful predictor of value, but its influence is mediated by channel, timing, and framing. By continuing to default to email-first pathways, uniform asks, and batch-driven timing, WVA risks undervaluing its most loyal supporters and exhausting its broader base.

1. Channel Misalignment

WVA's channel strategy prioritizes efficiency over resonance, defaulting to digital-heavy pathways even for high-value cohorts. This one-size approach risks undervaluing older and long-tenure supporters, who respond more strongly to tactile and conversational channels that build trust and reinforce relational cues (Bekkers & Wiepking, 2011). By treating all donors through the same email-led journeys, the campaign suppresses realized revenue and overlooks opportunities to align communication mode with supporter preferences.

2. Timing Misalignment with Donor Readiness

Donations peaked around Day 15, yet outbound activity clustered in Week 1 and after Day 18, reflecting production schedules rather than donor readiness. This mid-cycle peak aligns with pay cycles, DM response lag (10-14 days), and reinforcement from EDM reminders, making it the natural window of highest intent. As Shang & Croson (2009) note, donation intent follows salience curves rather than broadcast volume, meaning WVA must optimize cadence around this readiness curve to avoid both missing peak conversions and wasting late-cycle contacts.

3. Campaign Fatigue and Diminishing Returns

The sequencing analysis further shows that early touches such as Direct Mail and EDM1 generated donations, while late-stage touches like EDM3 and SMS2 delivered only \$0.8M despite heavy volume. This reflects diminishing marginal returns and aligns with sector evidence that repetitive, undifferentiated appeals can trigger donor fatigue, lowering both responsiveness and trust (Das, 2025). The implication is that WVA's broadcast-heavy strategy may prioritize volume over impact, inadvertently eroding supporter goodwill.

4. Campaign Dependence

Predictive results further demonstrates that channel effectiveness is not absolute but contingent on campaign context. Education appeals benefited from Direct Mail, which provided credibility and depth for abstract causes, whereas Christmas - competing for attention in a crowded season - drove stronger mid-tenure responses through quick digital channels like Email. Birthday campaigns, with their personal framing, aligned best with tactile prompts that reinforced emotional salience. This pattern reflects behavioral evidence that the interaction of message framing, donor identity, and delivery medium shapes giving outcomes (Winterich et al., 2013).

5. One-Size Ask Ladders Suppress Value

The lack of personalisation in WVA's ask strategy is problematic because donor behaviour is not uniform across age or tenure groups. Research shows that factors such as age, attitudes, and moral norms significantly shape donation intentions (White, 2023). Younger donors tend to give smaller, more impulsive gifts, while older and long-tenure supporters demonstrate higher capacity and commitment, often contributing larger amounts when engaged through trusted channels. By applying a single, flat ask ladder across all campaigns and supporter types, WVA overlooks these behavioural differences, which suppresses upgrades, risks overstretching entry-level donors, and underutilises the generosity of loyal, long-tenure supporters.

Recommendation

Recommendation 1. Channel Alignment

It is recommended for WVA to restructure channel allocation so that communication journeys better match donor profiles, shifting the balance between digital and tactile channels to maximise value while maintaining efficiency. Higher-value gifts consistently flow through Direct Mail and Telephone, while digital-first paths deliver lower returns, underscoring the need for a differentiated mix by age and tenure.

Age	Rebalance channel mix	Actions
+55(Tenure >10)	35-40% DM/Phone ≤50% EDM remainder SMS/Web	- Shift at least 20% of EDM1 volume into DM (+Coupon) - Use DM/phone for first & last touches - Replace EDM3/SMS2 with DM eminder or short call
< 45	≥70% Email/SMS; include 1 DM or Telephone per cycle (20-25%)	Digital remains core for efficiency, but one tactile/conversational touch (DM/Phone)

Measure of Success:

- Track average gift size growth among older cohorts (55+).
- Monitor revenue per 1,000 sends before and after reallocating late-stage touches.
- Assess unsubscribe/opt-out rates across younger donors to ensure no increase.

Business Impact: By aligning channels with supporter value, WVA reduces wasted EDM volume and redirects effort into the channels proven to drive higher-value gifts. This creates immediate financial lift (through higher average gift sizes) while also reducing long-term donor fatigue, protecting supporter trust, and improving campaign ROI without increasing overall contact volume.

Recommendation 2. Timing Optimization

BounceBack should align contact with donor readiness by focusing on the mid-month peak, keeping early touches strong, and transitioning later touches into lighter, emotionally framed communications. This ensures maximum conversion during Days 13-21 while still allowing incremental revenue beyond the third touch.

Time Window	Focus	Key Activities
Day 1-7	Awareness	Pre-notice email/SMS; DM prep; milestone reminder.
Day 8-12	Anticipation	Light nudges only; vendor confirms DM delivery.
Day 13-15	Conversion (Peak)	DM lands + paired EDM within 24h; strongest ask; send EDM/SMS in mornings.
Day 16-18	Reinforcement	SMS/EDM reminder only for engaged non-donors; urgency/emotional framing.
Day 19-21	Final Push	One reminder (DM/EDM/short call for high-value); round out 3 strong asks.
Day 21-30	Stewardship + Light Asks	Thank-you and impact comms; emotional storytelling; small triggered nudges (resend/abandoned) to capture late gifts.

Additional Rules

- Core cycle** = 3 strong broadcasts; post-Day 21 focus on stewardship and intent-triggered nudges.
- Segment pacing:** 55+ / tenure ≥10 yrs get DM + EDM (SMS only if engaged); <45s/short-tenure start with EDM/SMS and drop by Day 20 if inactive.

Measurement & Guardrail

- Track: revenue/1,000 sends, response rate, average gift, conversion latency, opt-outs.
- Daily cuts for Days 10-20. Success = ≥ 5 pp revenue share uplift before Day 21, stable average gift, opt-outs at/below control.
- Weekly review to decide stop/continue/scale; pause late broadcasts if guardrails breached.

Business Impact: These actions concentrate contact in the most productive window, eliminate low-yield activity later in the cycle and convert donor readiness into outcomes without additional spend while remaining feasible within the Bounceback cadence.

Recommendation 3. Ask Framing and Value

BounceBack should replace its uniform ask ladders with donation requests scaled to supporter capacity, reflecting tenure and past giving patterns. This ensures entry-level donors receive accessible asks while long-tenure supporters are offered aspirational yet realistic options.

Key Actions

- **Scale asks to donor capacity:** Anchor suggested amounts to a supporter's last gift and tenure band. Supporters with <40 years' tenure should be kept within accessible ranges (e.g., up to \$200), while those with >40 years' tenure should be offered higher bands (e.g., up to \$500) with a write-in option for further stretch.
- **A/B test ask framing:** Compare round-up (e.g., \$40–\$50), band-aligned (e.g., \$45 in a \$40–50 tier), and milestone-based asks (e.g., "\$50 funds a school term") to determine which style maximises conversion and upgrade.
- **Rotate BounceBack themes:** Cycle between Education, Birthday, and Christmas appeals to reduce fatigue and sustain donor engagement.
- **Simple routing rules:** In CRM/ESP, default older/long-tenure donors to Direct Mail with higher anchors, and younger/short-tenure donors to EDM/SMS with gentler ladders.

Measure of Success

- Track average gift size before and after tailored ladders are introduced.
- Monitor upgrade rates (supporters giving above their last gift) by tenure cohort.
- Compare conversion and donation values across different framing tests.

Business Impact: By tailoring asks to donor capacity, WVA reduces drop-offs from unrealistic anchors, encourages incremental upgrades among newer donors, and unlocks higher-value gifts from long-tenure supporters. This strengthens revenue efficiency while maintaining donor trust and sustaining long-term engagement.

Limitation & Future Improvement

While the analysis provides strong evidence for optimising the BounceBack campaign, two key limitations should be acknowledged:

- **Predictive Model Reliability**

The Random Forest models yielded a Kappa score of 0, meaning the models could distinguish Low donors but failed to reliably separate Medium and High donors. This reduces confidence in using the models for precise donor-level targeting. The implication is that predictive insights are best applied as decision-support tools to highlight broad patterns (e.g., tenure thresholds, channel effects) rather than as automated rules. Without further model refinement or additional predictor variables, relying on them for segmentation could risk misallocating high-value donors.

- **Data Consistency in Tenure Fields**

The supporter grouping tables revealed tenure inconsistencies, with fields such as child sponsor tenure, pledger tenure, and supporter tenure not aligning logically with age bands. This points to manual entry or static categorisation. Such inconsistencies can distort age-tenure interactions in the analysis, leading to misleading thresholds in decision trees or inflated importance of tenure as a predictor.

Mitigation: Tenure should be system-generated, calculated directly from core dates (e.g., date of birth, sponsorship start date, pledge start date). Automating tenure updates would ensure real-time accuracy, reduce data storage costs, and provide a stronger foundation for predictive modelling and segmentation.

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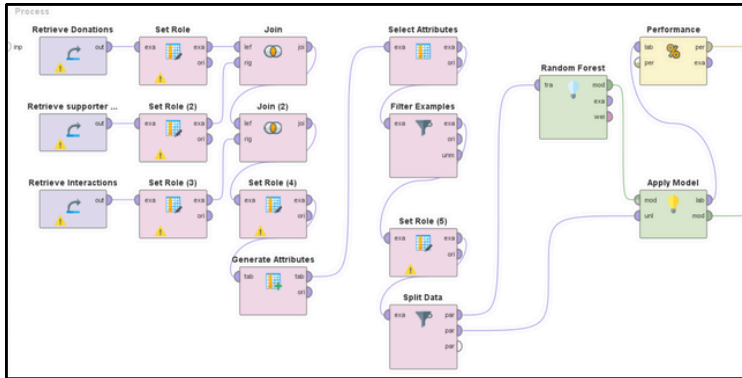
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Appendices

Appendix 1. Predictive Model Process



Parameters

Random Forest

number of trees: 100

criterion: gain ratio

maximal depth: 10

☒ apply pruning

confidence: 0.1

☒ apply prepruning

minimal gain: 0.01

minimal leaf size: 2

minimal size for split: 4

number of prepruning alternatives: 3

☐ random splits

☒ guess subset ratio

voting strategy: confidence v...

Parameters

Join

☒ remove double attributes

join type: inner

☐ use id attribute as key

key attributes: [Edit List \(2\)...](#)

☐ keep both join attributes

Select Attributes

Select Attributes: select subset

Select the attribute subset

Attributes:

Selected Attributes:

> Join Logic Explanation

- Identifiers:** Joins are made on Deakin_SupporterID (supporter) and Campaign_Key / DonatedCampaignKey (campaign).
- Relationships:** One supporter can be linked to many campaigns, and one campaign can involve many supporters → many-to-many.
- Linking Tables:** Donations, Supporter Grouping, and Interactions act as bridge tables capturing supporter–campaign pairs.
- Purpose:** This join merges donation details, supporter attributes, and interaction records into one dataset for modelling.

Appendix 2. Predictive Models Performance

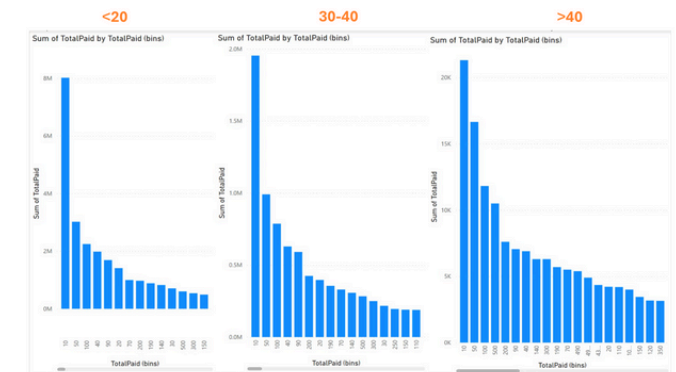
Birthday

accuracy: 79.78%					
	true Low	true Medium	true High	true No	class precision
pred Low	197029	51563	28482	1368	70.76%
pred Medium	0	0	0	0	0.00%
pred High	0	0	0	0	0.00%
pred No	0	0	0	0	0.00%
class recall	100.00%	0.00%	0.00%	0.00%	
classification_error: 29.24%					
	true Low	true Medium	true High	true No	class precision
pred Low	197029	51563	28482	1368	70.76%
pred Medium	0	0	0	0	0.00%
pred High	0	0	0	0	0.00%
pred No	0	0	0	0	0.00%
class recall	100.00%	0.00%	0.00%	0.00%	
kappa: 0.000					
	true Low	true Medium	true High	true No	class precision
pred Low	197029	51563	28482	1368	70.76%
pred Medium	0	0	0	0	0.00%
pred High	0	0	0	0	0.00%
pred No	0	0	0	0	0.00%
class recall	100.00%	0.00%	0.00%	0.00%	

Education

accuracy: 69.24%					
	true Low	true Medium	true High	true No	class precision
pred Low	132623	37786	19186	1934	69.24%
pred Medium	2	0	0	0	0.00%
pred High	0	0	0	0	0.00%
pred No	0	0	0	0	0.00%
class recall	100.00%	0.00%	0.00%	0.00%	
classification_error: 30.76%					
	true Low	true Medium	true High	true No	class precision
pred Low	132623	37786	19186	1934	69.24%
pred Medium	2	0	0	0	0.00%
pred High	0	0	0	0	0.00%
pred No	0	0	0	0	0.00%
class recall	100.00%	0.00%	0.00%	0.00%	
kappa: <0.000					
	true Low	true Medium	true High	true No	class precision
pred Low	132623	37786	19186	1934	69.24%
pred Medium	2	0	0	0	0.00%
pred High	0	0	0	0	0.00%
pred No	0	0	0	0	0.00%
class recall	100.00%	0.00%	0.00%	0.00%	

Appendix 3: Donation Paid Amounts for Different Tenures



Christmas

accuracy: 68.30%					
	true Low	true Medium	true High	true No	class precision
pred Low	247953	71089	41295	2286	68.30%
pred Medium	0	0	0	0	0.00%
pred High	0	0	0	0	0.00%
pred No	0	0	0	0	0.00%
class recall	100.00%	0.00%	0.00%	0.00%	
classification_error: 31.70%					
	true Low	true Medium	true High	true No	class precision
pred Low	247953	71089	41295	2286	68.30%
pred Medium	0	0	0	0	0.00%
pred High	0	0	0	0	0.00%
pred No	0	0	0	0	0.00%
class recall	100.00%	0.00%	0.00%	0.00%	
kappa: 0.000					
	true Low	true Medium	true High	true No	class precision
pred Low	247953	71089	41295	2286	68.30%
pred Medium	0	0	0	0	0.00%
pred High	0	0	0	0	0.00%
pred No	0	0	0	0	0.00%
class recall	100.00%	0.00%	0.00%	0.00%	